HARD · ADVANCED MATH

Nonlinear functions

01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

x	p(x)
−2	5
−1	0
0	−3
1	−1
2	0

The table above gives selected values of a polynomial function p. Based on the values in the table, which of the following must be a factor of p ?

A. $(x-3)$

B. $(x+3)$

C. $(x-1)(x+2)$

D. $(x+1)(x-2)$

Q2 ADVANCED MATH

The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

I. $fx = 330.4^{x+3}$
II. $gx = 330.160.4^{x-2}$

A. I only

B. II only

C. I and II

D. Neither I nor II

x	gx
-27	3
-9	0
21	5

The table shows three values of x and their corresponding values of gx, where $gx = \frac{fx}{x+3}$ and f is a linear function. What is the y-intercept of the graph of $y = fx$ in the xy-plane?

A. $0, 36$

B. $0, 12$

C. $0, 4$

D. $0, -9$

Q4 GRID-IN ADVANCED MATH

$$f x = x + 7^2 + 4$$

The function f is defined by the given equation. For what value of x does $f x$ reach its minimum?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN ADVANCED MATH

$$f x = x - 2x + 15$$

The function f is defined by the given equation. For what value of x does $f x$ reach its minimum?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6 ADVANCED MATH

For the exponential function f, the value of $f 1$ is k, where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

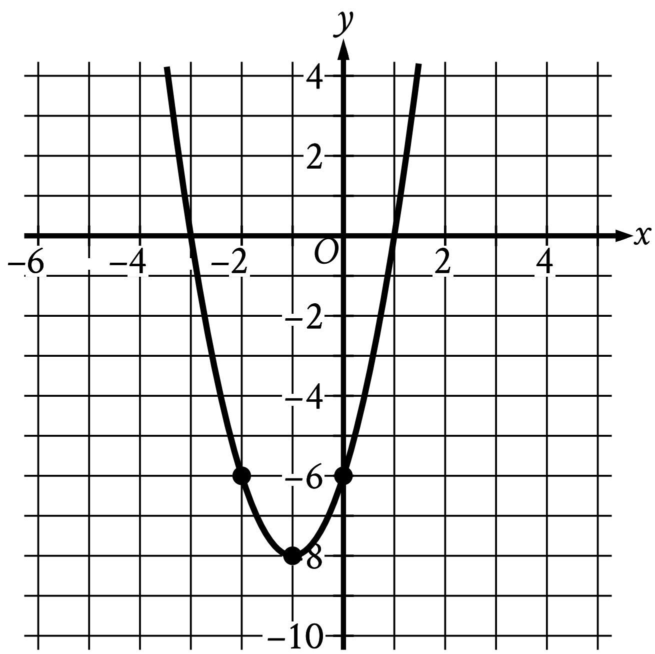
A. $f x = 502^{x + 1}$

B. $f x = 802^{x}$

C. $f x = 1282^{x - 1}$

D. $f x = 2052^{x - 2}$

Q7 GRID-IN ADVANCED MATH



- The parabola opens upward.
- The vertex is at point (negative 1 comma negative 8).
- The parabola passes through the following points:
  - (negative 2 comma negative 6)
  - (negative 1 comma negative 8)
  - (0 comma negative 6)

The graph of $y = 2x^2 + bx + c$ is shown, where b and c are constants. What is the value of bc?

STUDENT-PRODUCED RESPONSE

4 4 4 4
5 5 5 5
6 6 6 6
7 7 7 7
8 8 8 8
9 9 9 9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8 ADVANCED MATH

$$fx = 9(4)^{x}$$

The function f is defined by the given equation. If $g(x) = f(x+2)$, which of the following equations defines the function g?

A. $gx = 184^{x}$

B. $gx = 1444^{x}$

C. $gx = 188^{x}$

D. $gx = 8116^{x}$

Q9 ADVANCED MATH

At the time that an article was first featured on the home page of a news website, there were 40 comments on the article. An exponential model estimates that at the end of each hour after the article was first featured on the home page, the number of comments on the article had increased by 190% of the number of comments on the article at the end of the previous hour. Which of the following equations best represents this model, where C is the estimated number of comments on the article t hours after the article was first featured on the home page and $t \leq 4$?

A. $C = 401.19^t$

B. $C = 401.9^t$

C. $C = 4019^t$

D. $C = 402.9^t$

Q10 GRID-IN ADVANCED MATH

x	y
21	-8
23	8
25	-8

The table shows three values of x and their corresponding values of y, where $y = fx + 4$ and f is a quadratic function. What is the y-coordinate of the y-intercept of the graph of $y = fx$ in the xy-plane?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.